

EN 623: Solid Mechanics for Energy Systems

Course Code	EN 623
Course Name	Solid Mechanics for Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Mechanics of Solid
Intended for	UG/M.Tech/MS
Distribution	Compulsory for Energy Engineering (Mechanical) and Elective for others
Semester	Odd/Even
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Preamble

This course will possess the ability to apply the principles of solid mechanics to solve engineering problems and to design energy systems or components to meet desired needs; including (a) to idealize an energy system or component for the purposes of stress analysis; (b) to use appropriate numerical and analytical techniques to model the energy system.

Course Modules

- **Introduction:** Motivation for course; Review of elementary solid mechanics (1 contact hours)
- **Kinematics:** deformation, displacements, strain tensors, strain-displacement relations, compatibility equations (3 contact hours)
- **Kinetics:** traction vector, stress tensors, principle stresses, equations of motion, equilibrium (3 contact hours)
- **Constitutive Equations:** anisotropic/orthotropic/transversely isotropic/ isotropic materials, generalized Hooke's law, linearized elasticity (3 contact hours)
- **Basic Principles:** stress and displacement formulations for energy system, superposition, St. Venant's principle, boundary conditions (3 contact hours)
- **Variational Methods:** strain energy, uniqueness, complementary energy, potential energy, virtual work principle, minimum total potential/complementary energy principle, approximate solutions (4 contact hours)
- **Two-Dimensional (2-D) Theory:** plane stress and plane strain, anti-plane shear, Airy's stress function, inverse method, complex variable method, displacement solution (3 contact hours)
- **2-D Problems:** Energy component (beam) bending, plate with a hole, curved beams, pressurized cylinders, wedges, rotating disks (3 contact hours)
- **Torsion:** St. Venant torsion theory, Prandtl stress function, membrane analogy (3 contact hours)
- **The finite element method for numerical analyses of energy system:** Finite elements, Element, interpolation functions, Element strains, stresses and strain energy density, Element Stiffness Matrix, Global Stiffness Matrix, Boundary conditions. Demonstration of FEM based software (ANSYS) for analyzing stress, deformation and failure in components, assemblies and structures. (8 contact hours)
- **Boundary value problems for energy system made of elastic-plastic materials:** Tension-torsion of thin walled tubes, Plastic limit load, Approximate methods in metal forming (4 contact hours)
- **Failure modes in Energy components:** Fracture, Fatigue, Buckling, Large deflections, Plastic collapse. (4 contact hours)

References

1. S. P. Timoshenko and J. N. Goodier, *Theory of Elasticity*, 3rd edition, McGraw-Hill, New York, 1970
2. W. S. Slaughter, *The Linearized Theory of Elasticity*, Birkhäuser, Boston, 2002
3. R. W. Little, *Elasticity*, Prentice-Hall, Englewood Cliffs, New Jersey, 1973
4. P. Boresi and K. P. Chong, *Elasticity in Engineering Mechanics*, 2nd edition, Wiley, New York, 2000
5. Hertzberg R.W., Vinci R.P., Hertzberg J., *Deformation and Fracture Mechanics of Engineering Materials*, Wiley, 2012